

The empirical recurrence for OEIS A231215, recovered from its tail

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Abstract

OEIS A231215 records “Empirical recurrence of order 68” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 2500$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 69$ — every index at which it can be written down. Its last coefficient is $c_{68} = -7572096 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A231215 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with no element unequal to a strict majority of its horizontal and vertical neighbors, with values 0..2 introduced in row major order.”. It has offset 1 and begins

12, 59, 408, 2892, 20027, 139438, 969461, 6745110, ...

The entry states:

Empirical recurrence of order 68 (see link above).

As of the “Last modified” line on the live entry (Nov 01 2014 03:11:38, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 2500$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2500$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 5000$ exact terms from $n = 1$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	20	c_{18}	−34679131	c_{35}	20267483201	c_{52}	−799032802
c_2	−154	c_{19}	116011810	c_{36}	−21759800554	c_{53}	−9583393560
c_3	520	c_{20}	−205261369	c_{37}	10591628217	c_{54}	−10648011741
c_4	11	c_{21}	143858615	c_{38}	11097988443	c_{55}	−7377316040
c_5	−7095	c_{22}	124807534	c_{39}	−21903698601	c_{56}	−5023560375
c_6	27961	c_{23}	−385517779	c_{40}	17900969644	c_{57}	1358660487
c_7	−38414	c_{24}	428809089	c_{41}	−6424273044	c_{58}	1915160266
c_8	−69243	c_{25}	−516139870	c_{42}	−4717811421	c_{59}	3991732301
c_9	398256	c_{26}	1185784733	c_{43}	14616128287	c_{60}	2626969675
c_{10}	−623012	c_{27}	−2330766318	c_{44}	−31505016012	c_{61}	1462351138
c_{11}	−306046	c_{28}	3245592299	c_{45}	16667025671	c_{62}	655005218
c_{12}	2829850	c_{29}	−3729842968	c_{46}	−4921339559	c_{63}	−129004018
c_{13}	−3927866	c_{30}	4128660160	c_{47}	−10821272226	c_{64}	−120743732
c_{14}	−1924374	c_{31}	−3898932511	c_{48}	22347762692	c_{65}	−147195984
c_{15}	13843441	c_{32}	1102061485	c_{49}	−492504540	c_{66}	−67840944
c_{16}	−19547041	c_{33}	4565983775	c_{50}	18058388211	c_{67}	−19310400
c_{17}	15384060	c_{34}	−12478881886	c_{51}	1443855657	c_{68}	−7572096

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 69$, and it is the recurrence OEIS A231215 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{67} - \dots - c_{68}$. Its constant term is $-c_{68} = 7572096$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From

$\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -7572096 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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